



In-season estimates of ad-intact fish returning to the Krassel PIT-array

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Summary

- Current PIT-tag expansions at the Krasel PIT-array (KRA; as of October 17th, 2023)¹
 - 656 natural-origin fish estimated from 118 observed tags and 18% sampling/tagging rate (Tables 1 and 2)
 - Using a 20% tagging rate the estimate is 590.
- KRS was non-operational: 7/10, 7/11, 7/14, 7/15, and 7/16
 - Natural-origin PIT-tag passage during this period is estimated as ≤ 10 tags for an additional 56 natural-origin fish.
- The average passage date of $\geq 75\%$ of the total return across KRS is July 15th (Figure ??).
- End of season natural-origin estimate: 948
 - NOR: $(656 + 56)/.75 = 949$
- Current Lower Granite Estimate of McCall Integrated fish is 465.
 - Estimated at KRS after accounting for SbyC is 443 (95% conversion)
- **Total ad-intact fish at KRS: 1,413**
 - $949 \text{ (NOR)} + 465 \text{ (Integrated)} = 1,414$

¹Scripts to reproduce these summaries are available at: "https://github.com/NPTfisheries/sf_salmon_harvest.com".

Assumptions

- 100% conversion of integrated fish to KRS.
- 100% detection efficiency at KRS
 - In 2021 and 2022 KRS detections efficiency was $\geq 95\%$.
- Tagging rate at LGR of 18%.
 - Only 4 out of 118 NOR tags were marked at LGR during the 29% sampling rate, all others marked at 18%.
- Static interpolation rate of 2 tags during the missed period at KRS when observations are typically decreasing.

Table 1: *Current tag expansion rates and estimates for SFSR release groups at KRS.*

Release Year	Group	SbyC	Expansion	n_tags	Estimate
2020	Integrated	RAL	6.710924	1	7
2020	Integrated	RTR	1.000000	1	1
2020	Segregated	RAL	50.028692	1	50
2021	Integrated	RAL	6.995739	56	392
2021	Integrated	RTR	1.000000	24	24
2021	Segregated	RAL	50.761811	50	2,538
2021	Segregated	RTR	1.000000	28	28
2022	Integrated	RAL	8.312441	12	100
2022	Integrated	RTR	1.000000	3	3
2022	Integrated			4	
2022	Segregated	RAL	39.299292	10	393
2022	Segregated	RTR	1.000000	5	5
2022	Segregated			1	
2023	NOR		5.555556	154	856

Table 2: *Current SFSR release group estimates at KRS.*

Group	n_tags	Estimate
Integrated	101	527
NOR	154	856
Segregated	95	3,014

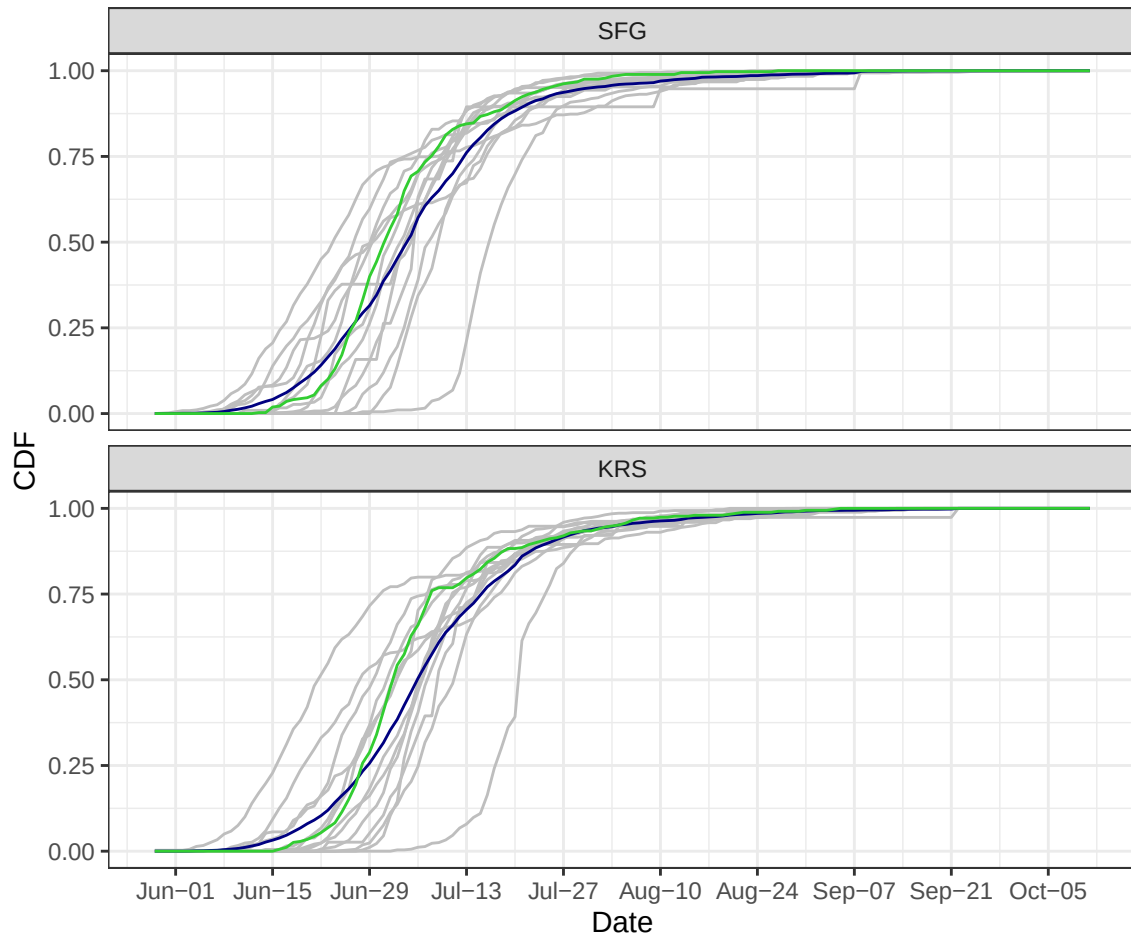


Figure 1: *Run-timing of SFSR hatchery (segregated and integrated) and natural origin returns across PIT-arrays. The grey lines indicate individual spawn year returns for 2010-2022, the dark blue line is the average cumulative proportion of returns across the same years, and the green line represents the current 2023 return.*